Version 1
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Validation of SOFA-2 score – synopsis

Background

The Sequential Organ Failure Assessment (SOFA) score was first introduced in 1996 to assess and evaluate organ dysfunction in critically ill patients[1] and is currently used as a key criterion in the diagnosis of sepsis[2]. Meanwhile intensive care medicine has evolved substantially, with several interventions being replaced or discontinued and new procedures, drugs, devices and monitoring strategies being introduced[3]. As a result, experts have argued that the current SOFA score is outdated, as it fails to capture modern interventions used to support failing organs[4,5]. This led to the development of the SOFA-2 score, which consisted of a 8-stage process including a modified Delphi process and an internal and external validation[6,7].

The SOFA-2 score has kept the overall structure of the original score, including the same six organ systems with a total score ranging from 0 to 24. Although the inclusion of two additional organ systems, the immune and gastrointestinal systems, was initially proposed, these were ultimately not incorporated following further evaluation. The primary modifications in SOFA-2 therefore consist of updated laboratory thresholds and the inclusion of commonly used drugs and modern mechanical organ support modalities. The study found similar predictive validity of SOFA-1 and SOFA-2 for ICU mortality when using area under the receiver operating characteristics (AUROC) curve[6].

The current validation of the SOFA-2 score is based exclusively on critically ill patients admitted to intensive care units (ICU). Consequently, it remains unclear whether the updated scoring system is applicable to a broader, undifferentiated population of patients who do not require intensive care treatment. Given the importance of SOFA in the identification and diagnosis of sepsis, it is considered especially essential to investigate whether SOFA-2 modifies the assessment of patients with acute infection. With this study we therefore aim to validate the SOFA-2 score in adult patients admitted to emergency departments (ED) with suspected infection by comparing the distribution of scores and the predictive validity of the original SOFA.

Methods

Study population

This retrospective multicenter cohort study uses data from all admissions to emergency departments (ED) in Eastern Denmark from January 1, 2018, to January 31, 2023.

Inclusion criteria:

1. Age ≥ 18 years
2. Antibiotic treatment or collection of any specimen for microbiological isolation within the first 72 hours of hospital admission

The following additional inclusion criterion will be used for sensitivity analyses (termed "the ICU cohort"):

3. Admission to ICU during the first 24 hours of hospital admission

Objectives

Primary objective:

- Changes in patients' Sequential Organ Failure Assessment (SOFA) score after the implementation of SOFA-2 compared with SOFA-1.

Secondary objective:

- Predictive validity of SOFA-1 and SOFA-2 for 30-day mortality and ICU-admission

Study design

This retrospective multicenter cohort study will be reported according to Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.

Sample size

No formal sample size can be performed, as the primary objective is to validate SOFA-2 in an as large population as possible.

Statistical principles

P-values below 0.05 are considered statistically significant. Ninety-five percent confidence intervals are presented where feasible. No correction for multiple comparison is performed, as all study results are considered exploratory.

Data extraction

To evaluate the validity of the SOFA-2, we will compare it to the original SOFA as it was presented in the original publication, as well as the original SOFA score with common simplifications, which allows use in non-ICU settings.

Data for constructing the SOFA-1 and -2 scores will be extracted from the electronic health record (EHR) and will be of the type "real-world data", i.e., data which is recorded for the purpose of diagnosis and treatment, and not for research or quality improvement purposes.

The following variables will be extracted from the EHR for constructing SOFA scores from admission to day seven:

- Glasgow Coma Scale (GCS)
- Alert, Verbal, Pain, Unresponsive (AVPU)
- Administration of Haloperidol or Olanzapine
- Peripheral oxygen saturation (SpO2)
- Arterial partial pressure of oxygen (PaO2)
- Fraction of inspired oxygen (FiO2)
- Supplemental oxygen
- High-flow nasal oxygen therapy (HFNO)
- Continuous positive airway pressure (CPAP)
- Non-invasive ventilation (NIV)
- Invasive ventilation
- Extra-corporeal membrane oxygenation (ECMO)
- Blood pressure (systolic, diastolic, and mean arterial pressures)
- Doses and duration of vasopressors and inotropes (dobutamine, dopamine, vasopressin, levosimendan, milrinone, phenylephrine, ephedrine)
- Bilirubin levels
- Creatinine levels
- Urine output
- Potassium levels
- pH levels
- Bicarbonate levels
- Use of renal replacement therapy (RRT)

- Platelet counts

SOFA scores will be calculated for each 24 hours of admission (with day one starting at admission, day two starting after 24 hours, etc.). The worst value from each day will be used for calculating the total score.

Details on how each component of the SOFA scores is calculated is presented in the appendix.

Statistical analysis

Primary objective

The primary objective, which is to compare the original SOFA to SOFA-2, will be assessed by descriptive statistics. The number (percentage) of patients assigned each score from 0-24 will be presented for both the original SOFA and SOFA-2. Further, the number (percentage) of patients assigned each score from 0-4 in each of the six domains will be presented for both scoring systems.

The frequency of patients being reclassified to a lower, the same, and a higher SOFA-2 value, compared to their original SOFA value, will be presented.

The primary objective will be assessed in two cohorts: patients admitted to the ICU during the first 24 hours of admission and all patients.

Secondary objectives

The first secondary objective, to assess the predictive validity of SOFA-2 for 30-day mortality and ICU admission, will be assessed using logistic regression. Results will be reported as odds ratios (OR) with 95% confidence intervals (CI).

Decision curve analysis (DCA) will be performed for both outcomes, comparing SOFA-2 to the original SOFA.

Area under the curve (AUC) for the receiver operating characteristic (ROC) will be estimated for both SOFA-2 and the original SOFA. Standard errors will be estimated using the DeLong method. Due to expected imbalances, the AUC for the precision-recall curve will also be estimated for both SOFA-2 and the original SOFA.

Thirty-day mortality will be assessed in both cohorts. ICU admission will only be assessed in all patients and not in the ICU cohort.

Missing data

Missing values will be handled by normal value imputation, which was the strategy used in the SOFA-2 development and validation article. This means that any patients with missing data for the variables necessary to calculate one or more of the six domains will receive a score of zero for that/those domain(s).

Sensitivity analyses

Sensitivity analyses will be performed. First, the original SOFA will be recalculated based on commonly used simplifications (see Appendix), and the primary analyses will be repeated comparing SOFA-2 to this score. Second, the primary analyses will be repeated using complete case data.

References

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Appendix

Brain

The *brain* component will be calculated accordingly:

	SOFA 2	**SOFA 1**	**SOFA 1 (pragmatic)**
0	1. GCS 15 or 2. AVPU "A" and GCS missing	GCS 15	1. GCS 15 or 2. AVPU "A" and GCS missing
1	1. GCS 13-14 or 2. administration of haloperidol/olanzapine and a. GCS 15 or b. GCS missing and AVPU "A" or c. GCS missing and AVPU missing or 3. AVPU "V"	GCS 13-14	1. GCS 13-14 or 2. AVPU "V"
2	1. GCS 9-12 or 2. AVPU "P"	GCS 10-12	1. GCS 10-12 or 2. AVPU "P"
3	1. GCS 6-8	GCS 6-9	1. GCS 6-9
4	1. GCS 3-5 or 2. AVPU "U"	GCS 3-5	1. GCS 3-5 or 2. AVPU "U"

Respiratory

The *respiratory* component will be calculated accordingly:

	SOFA 2	SOFA 1	SOFA 1 (pragmatic)
0	1. PaO2:FiO2 ratio >300 or 2. SpO2:FiO2 ratio >300	PaO2:FiO2 ratio ≥400	1. PaO2:FiO2 ratio ≥400 or 2. SpO2 ≥95%
1	1. PaO2:FiO2 ratio ≤300 or 2. SpO2:FiO2 ratio ≤300	PaO2:FiO2 ratio <400	1. PaO2:FiO2 ratio <400 or 2. SpO2 <95%
2	1. PaO2:FiO2 ratio ≤225 or 2. SpO2:FiO2 ratio ≤250	PaO2:FiO2 ratio <300	1. PaO2:FiO2 ratio <300 or 2. SpO2 <91%
3	1. Ventilatory support and a. PaO2:FiO2 ratio ≤150 or b. SpO2:FiO2 ratio ≤200	Mechanical ventilation and PaO2:FiO2 ratio <200	Mechanical ventilation and PaO2:FiO2 ratio <200
4	1. Ventilatory support or ECMO and a. PaO2:FiO2 ratio ≤75 or b. SpO2:FiO2 ratio ≤120	Mechanical ventilation and PaO2:FiO2 ratio <100	Mechanical ventilation and PaO2:FiO2 ratio <100
Only registered FiO2 values will be used, i.e., no estimations will be made based on supplemental oxygen. Patients without a registered FiO2 value will receive a value of 0.21. **Ventilatory support = high flow nasal oxygen, continuous positive airway pressure, non-invasive ventilation, invasive ventilation.** **Mechanical ventilation = continuous positive airway pressure, non-invasive ventilation, invasive ventilation.**			

Cardiovascular

The *cardiovascular* component will be calculated accordingly.

	SOFA 2	**SOFA 1**
0	MAP ≥70 mmHg	MAP ≥70 mmHg
1	MAP <70 mmHg	MAP <70 mmHg
2	1. Sum of norepinephrine and epinephrine ≤0.2 µg/kg/min or 2. Any dose of any other vasopressor or inotrope or 3. MAP <60 mmHg and no vasopressor or inotrope	1. Dopamine ≤5 µg/kg/min or 2. Dobutamine (any dose)
3	1. Sum of norepinephrine and epinephrine from >0.2 to ≤0.4 µg/kg/min or 2. Sum of norepinephrine and epinephrine ≤0.2 µg/kg/min and any dose of any other vasopressor or inotrope or 3. Dopamine from >20 to ≤40 µg/kg/min or 4. MAP <50 mmHg and no vasopressor or inotrope	1. Dopamine >5 µg/kg/min or 2. Epinephrine ≤0.1 µg/kg/min or 3. Norepinephrine ≤0.1 µg/kg/min
4	1. Sum of norepinephrine and epinephrine ≥0.4 µg/kg/min or 2. Sum of norepinephrine and epinephrine from >0.2 to ≤0.4 µg/kg/min and any other vasopressor or inotrope or 3. Dopamine >40 µg/kg/min or 4. Venoarterial extracorporeal membrane oxygenation, intra-aortic	1. Dopamine >15 µg/kg/min or 2. Epinephrine >0.1 µg/kg/min or 3. Norepinephrine >0.1 µg/kg/min

	balloon pump or left-ventricular assist device or 5. MAP <40 mmHg and no vasopressor or inotrope	

MAP is calculated as 1/3 * systolic blood pressure + 2/3 * diastolic blood pressure. Only MAP thresholds met at least twice with at least one hour between measurements will be considered valid.

Doses registered in non-weight-based units (i.e., ml/h and similar) will be set to the lowest possible dose.

Any other vasopressor or inotrope include: dobutamine, dopamine, vasopressin, levosimendan, milrinone, phenylephrine, ephedrine.

Liver

The *liver* component is calculated accordingly.

	SOFA 2	**SOFA 1**
0	Bilirubin ≤20.6 µmol/L	Bilirubin <20.0 µmol/L
1	Bilirubin ≤51.3 µmol/L	Bilirubin <33 µmol/L
2	Bilirubin ≤102.6 µmol/L	Bilirubin <102 µmol/L
3	Bilirubin ≤205.0 µmol/L	Bilirubin ≤204 µmol/L
4	Bilirubin >205.0 µmol/L	Bilirubin >204 µmol/L
Patients with a recorded ICD-10 code corresponding to chronic liver disease will only be allowed to score 0, 3 or 4.		

Kidney

The *kidney* component is calculated accordingly.

	SOFA 2	**SOFA 1**
0	Creatinine ≤110 µmol/L	Creatinine <110 µmol/L
1	1. Creatinine ≤170 µmol/L or 2. Urinary output <0.5 mL/kg/h for 6-12 h	Creatinine <171 µmol/L
2	1. Creatinine ≤300 µmol/L or 2. Urinary output <0.5 mL/kg/h for ≥12 h	Creatinine <300 µmol/L
3	1. Creatinine >300 µmol/L or 2. Urinary output <0.3 mL/kg/h for ≥24 h or 3. Anuria for ≥12 h	1. Creatinine ≤440 µmol/L or 2. Urinary output <500 mL/day
4	1. Receiving renal replacement therapy (excluding renal replacement therapy due to overdose) or 2. Creatinine >110 µmol/L or urinary output <0.3 mL/kg/h for ≥6 h and a. Potassium ≥6.0 mmol/L or b. pH ≤7.2 or c. Bicarbonate ≤12 mmol/L	1. Creatinine >440 µmol/L or 2. Urinary output <200 mL/day
Overdose is defined as having an international classification of diseases, tenth revision (ICD-10) code suggesting overdose (T36-T50) during the same admission. **Patients with a recorded ICD-10 code corresponding to chronic kidney disease will only be allowed to score 0, 3 or 4.**		

Hemostasis

The *hemostasis* component will be calculated accordingly:

	SOFA 2	**SOFA 1**
0	Platelets >150 * 10^3/µL	Platelets ≥150 * 10^3/µL
1	Platelets ≤150 * 10^3/µL	Platelets <150 * 10^3/µL
2	Platelets ≤100 * 10^3/µL	Platelets <100 * 10^3/µL
3	Platelets ≤80 * 10^3/µL	Platelets <50 * 10^3/µL
4	Platelets ≤50 * 10^3/µL	Platelets <20 * 10^3/µL